

**FIFTH SEMESTER DIPLOMA EXAMINATION IN MECHANICAL
ENGINEERING – MODEL QUESTION PAPER**

TED (10)5017

POWERPLANT ENGINEERING

Time:3 Hrs.

[maximum marks:100]

PART-A

I.(Answer all questions in one or two sentences. Each question carries 2 marks.)

- a. Define a boiler.
- b. List the advantages of steam turbines over steam engines.
- c. List the types of gas turbines.
- d. What are the principal parts of a reactor.
- e. What is dryness fraction of dry steam. 5*2=10 marks

PART-B

II. (Answer any five questions from the following. Each question carries 6 marks.)

1. Distinguish between wet steam, dry steam and super heated steam.
2. Classify the steam Nozzles and explain.
3. Define vacuum efficiency and list the factors affecting the vacuum efficiency.
4. Explain with the diagrams, the working of a constant pressure gas turbine (Closed type).
5. Compare Gas turbines with steam engines.
6. Explain nuclear reaction, fission, fusion and chain reaction
7. Explain with simple sketches the working of solar grain drier and solar cooker. 5*6=30 marks

PART-C

UNIT 1

III. a) Explain with the line diagram, the construction and working of Nestler boiler. -7 marks

b) Calculate the enthalpy of 1 kg of steam at a pressure of 8 bar and dryness fraction of 0.8. How much heat would be required to raise 2 kg of this steam from water at 20° C. -8 marks

OR

IV a) What is the use of boiler draught ?. Explain the function of natural draught and induced draught with simple sketches. -7 marks

b) Using mollier chart, find enthalpy drop and final condition of steam when it is expanded isentropically from and initial pressure of 30 bar and 350 ° C to a pressure of 1 bar. -8 marks

UNIT II

V a) Derive an expression for velocity of steam leaving a nozzle. -7 marks

b) State the purpose of compounding steam turbines .Explain pressure compounding and velocity compounding in steam turbines. -8 marks

OR

VI a) Briefly explain the working principle of double acting steam engine with simple line sketch. -7 marks

B) Define vacuum efficiency. Calculate the vacuum efficiency from the following data.

Vacuum at steam inlet to condenser is 700 mm of mercury; Barometric reading is 760 mm of mercury and Hot well temperature is 30 ° C. -8 marks

UNIT III

VII a) compare gas turbines with steam turbines. -7 marks

b) Explain the working of diesel power plant with block diagrams. -8 marks

OR

VIII a) Explain the principles of rocket propulsion ? -7 marks

B) Explain with flow diagrams and T – S diagram, the working of constant pressure open type gas turbine. -8 marks

UNIT IV

IX a) Explain the principle of working of biogas plant with sketch. -7 marks

b) Briefly describe the following

i) Nuclear reactors. ii) Moderators

iii) coolants iv) Control rods -8 marks

OR

X a) Identify the renewable sources of energy and briefly explain. -7 marks

b) Explain the working of a nuclear power plant with the help of a schematic diagram. -8 marks